The first-order κ_3 question

A controlled correction to the half-carried K×C terms
Research response | 10 October 2026

[D] The known coefficient deficit is real, but the reported full-restoration run does not isolate it. Scaling both feedback legs also scales their K×K product quadratically. I would first restore the K×C coefficient while holding that product fixed. This uses the existing sources and needs **no new dense leg or matrix product**. It is a concrete, cheap correction to test, not a demonstrated explanation of the entire error.

[S] I cannot identify the dominant remaining cause from the supplied moments. In particular, the leading $\kappa_4 \to \kappa_3$ response needs $\kappa_4(i,i,j,k)$, which the listed teachers do not contain. The candidate called "missing cubic response" also overlaps the already identified K×C terms; the three hypotheses are not independent.

Labels: [D] derived; [E] implementation estimate; [S] speculative interpretation or prediction; [Q] supplied observation or read-only source-code fact. All displayed equations and their constants are [D]. No network code, Monte Carlo, or accuracy test was run. Document generation and arithmetic accounting are not network experiments.

Separate cumulant order, Hermite order and covariance order

[D] All quantities below are *pre-ReLU*. Put $u=z-\mu$, $v_i=C_{ii}$, $T=\kappa_3(u)$, $Q=\kappa_4(u)$, and $R=\kappa_5(u)$. With $d=\Phi(\alpha)$, $e=\varphi(\alpha)/\sigma$ and $f=-\alpha\varphi(\alpha)/\sigma^2$, the cubic Hermite approximation is:

```
h_i = constant + d_i u_i + (e_i/2)(u_i^2-v_i)
                 + (f_i/6)(u_i^3-3v_i u_i) + ...
```

[D] For distinct i,j,k, the following cumulant identities are exact for a centered input with the required moments. They do not assume a Gaussian input:

```
cum(u_i^2-v_i, u_j, u_k)
  = Q_iijk + 2 C_ij C_ik
cum(u_i^3-3v_i u_i, u_j, u_k)
  = R_iiijk + 3 C_ij T_iik + 3 C_ik T_iij
cum(u_i^2-v_i, u_j^2-v_j, u_k)
  = R_iijjk + 4 C_ij T_ijk
    + 2 C_ik T_ijj + 2 C_jk T_iij
```

[D] Multiply the first identity by $e_i d_j d_k/2$ and sum the three placements. The second has coefficient $f_i d_j d_k/6$; the third $e_i e_j d_k/4$, summed over the three unordered quadratic pairs. Thus:

```
Delta_Q Tpost_ijk
  = (1/2) sum_cyclic e_i d_j d_k Q_iijk
```

[D] The cubic-linear-linear term already contains one of your K×C families. The quadratic-quadratic-linear term contains the other, plus $C_{ij}T_{ijk}$ transport curvature. The latter belongs to the separate C·T thread and is not proposed here. The R terms are additional missing information. They are first order in a formal non-Gaussian perturbation if $R=O(\varepsilon)$; calling them higher order requires an additional width-scaling argument.

[D] Purely Gaussian higher-Hermite diagrams, such as the covariance triangle and $C^2\times C$ double-edge terms, are zeroth order in non-Gaussianity. A Hermite-truncation experiment changes many diagrams at once and cannot, by itself, identify which cumulant-recursion term is wrong.

Ranking and the scaling confound

Prioritize what is identifiable; do not turn a code fact into an accuracy claim

Priority	Assessment
1. K×C coefficient deficit	[D] An explicitly known first-order omission. [S] Highest-priority controlled test because it is identifiable and cheap; not proof it dominates the remaining error.
2. Mixed κ_4 feedback	[D] Q_iijk enters directly at first order. [S] A strong candidate for the unexplained remainder, but its magnitude and direction are unobserved here. A κ_4-diagonal oracle does not test this channel.
3. An independent “cubic birth”	[D] Partly the same mechanism as priority 1. [S] Residual cubic/higher-Hermite effects remain possible, but the supplied Hermite table cannot separate them from Q, R or input-state error.

[Q] The reported error growth, source-tier ablation, oracle gains and feedback-scale outcomes motivate this order of investigation. [D] None uniquely identifies a missing tensor sector. Random-looking slice error with near-unit slope is compatible with several anisotropic errors after weight mixing, including errors in the covariance used to create each source.

The current source contains a quadratic cross-term

[D] Freeze a birth's inputs and feedback factors. Write X and Y for the feedback matrices at unit feedback scale, B=A diag(w_2), and define Sym as the average over permutations, summed over hub columns. The first source block is:

```
H(beta) = Sym(3A + beta X, P, B + beta Y)
H0 = Sym(3A, P, B)
H1 = Sym(X, P, B) + 3 Sym(A, P, Y)
H2 = Sym(X, P, Y)
H(beta) = H0 + beta H1 + beta^2 H2
```

[D] X,Y are each linear in the represented incoming κ_3 slice, at fixed compression factors. H1 is the first-order feedback; H2 is quadratic. If β_0=1.25 and β=2 [Q], the source change is:

```
H(2)-H(1.25) = 0.75 H1 + 2.4375 H2
```

[Q-code] In `submissions/v55As125/estimator.py`, lines 1542 and 1544 multiply both right feedback factors by FB_SCALE. Lines 1555-1563 include their product in the birth slices. The source readout likewise contains Xt*Yt. This confirms the algebraic confound in that implementation; the source code was read, not executed.

[S] This might explain why full restoration hurts, but it is not established. [D] Even if the H2 effect is negligible, removing a real κ_3 error can worsen final means through cancellation with covariance and κ_4 errors. For final residual r and induced mean correction p, the change in squared error is $2\eta(r\cdot p)+\eta^2||p||^2$. Improving a tensor norm does not determine the sign of that expression.

[D] The quoted small K×K terms therefore do not prove that H2 caused the paradox; they also do not justify ignoring the scale confound. The controlled test below isolates it. [S] If this test fails, move mixed κ_4 feedback to the front of the investigation instead of repeatedly retuning the same coefficient.

One correction: restore only the linear part

Same transported legs and full source history; modified contractions

[D] Replace the block interpretation by $H^*=H0+\beta H1+\beta_0^2 H2$. Set $\beta=2$ for the restored first-order coefficient and keep the chosen baseline β_0 fixed. Equivalently, using the existing scaled feedback matrices $X\beta=\beta X$ and $Y\beta=\beta Y$:

```
gamma = (beta0/beta)^2
H* = Sym(3A, P, B) + Sym(Xbeta, P, B)
     + 3 Sym(A, P, Ybeta)
     + gamma Sym(Xbeta, P, Ybeta)
```

[D] For $\beta_0=1.25$, $\gamma=25/64$; if the reference run used $\beta_0=1$, use $\gamma=1/4$. This preserves the baseline quadratic term at fixed birth inputs. It is not a complete second-order Edgeworth closure. The restoration is exact for the represented feedback factors; their existing rank approximation is still present.

State and transport

[D] Keep exactly the existing A,P [n,n], F_1,F_2 [n,r], R_1,R_2 [n,r] with $X\beta=F_1R_1^T$ and $Y\beta=F_2R_2^T$, plus γ per birth if the scale varies by birth. Every live leg still evolves by $W^T\mathrm{diag}(d)$. All four polynomial blocks therefore remain represented for the full lifetime of the source. Do not drop the correction when a source enters the old tier.

Readout formulas that make the patch unambiguous

[D] The existing readout already computes the cross-block contribution below. Multiply only this contribution by γ; leave the Gaussian and the two linear-feedback contributions at their new β scale:

```
D21_cross = [(Xbeta*P) Ybeta.T
             + (Xbeta*Ybeta) P.T
             + (P*Ybeta) Xbeta.T] / 3
d3_cross = rowsum(Xbeta*P*Ybeta)
```

[D] This is a coefficient change inside the existing contractions, not a request for three extra dense products. In the fused source kernel: replace the coefficient of Xt*Yt in LP and d3 by γ times its old value; replace Xt*P/3 inside GY by γ Xt*P/3; and replace P*Yt inside GX by γ P*Yt. The linear LA terms stay unchanged. Apply the same algebra in dense, shared-basis and diagonal-only paths.

Also correct the newborn repeated-index compensation

[D] At birth P=I. Put x=diag(Xβ), y=diag(Yβ), and let S_{naive} be the birth D21 of the newly scaled B1 block. Before the existing residual/M-leg construction, use:

```
S_cross = [x[:,None]*Ybeta.T + Xbeta*Ybeta
           + y[:,None]*Xbeta.T] / 3
S_birth = S_naive - (1-gamma) S_cross
d3_birth_B1 = gamma*x*y
```

[D] Apply this only to B1 before adding the other birth blocks, then run the usual compensation against the univariate/bivariate post-ReLU closure. Otherwise the patch changes repeated-index moments that were supposed to stay fixed. Existing residual-rank compression may still approximate this compensation; test the identity before compression when auditing it.

[D] No new dense transport is needed and no new tensor family is introduced. [S] This is materially different from merely adding feedback rank, or rescaling both feedback factors and accepting their new product. No accuracy improvement is claimed until the controlled test passes.

Cost and a stored-data falsifier

Full-width upper bound for the explicit fused patch; no benchmark claims

[D] Let n=1024 [Q], S=120 source-layer readouts and B=15 births in the depth-16 network [D]. Counting every source readout as full D21 is conservative because the final layer needs only d3. Scalar coefficients replace constants in existing multiplies. One additional n×n multiply in GX suffices per source readout. The explicit birth subtraction above uses seven n×n elementwise operations and one length-n multiply.

Additional work [D]	Scalar FLOPs	Primitive calls
Source readouts	$S\,n^2$	S
Birth compensation	$B(7n^2+n)$	8B
Total reference patch	235,944,960 = 0.109870434 u	240
New dense matrix products	0	0

[D] This count is independent of feedback rank: the same products are already present. It assumes coefficients are changed inside the shown fused contractions, not recomputed as separate counter-tensors. Added persistent state is at most one scalar per birth; one n×n scratch buffer is 4 MiB in float32, and an existing scratch buffer can be reused.

[E] Meter-specific copies, allocation charges and the exact shared-basis implementation have not been benchmarked. Even allowing several additional pointwise passes per source gives substantial room below +30 u / +3,000 operations [Q]. The table is an implementation-specific derived ledger, not a metered run.

Falsify the scale-confound explanation first

[D] Start at a stored late capture, e.g. L14→L15, then repeat at L3/7/11/15→the next layer. Freeze the captured μ,C, incoming source tensor, feedback compression factors and post-ReLU slice-closure outputs. Rebuild only the newborn block and its compensation in the following three variants:

Variant	Linear H1 coefficient	Quadratic H2 coefficient
Baseline	β_0	β_0^2
Naive restoration	2	4
Proposed restoration	2	β_0^2

[D] Transport these births through the stored next W and compare their next D21/d3 to the same next-layer teacher. Keep compensation exact/dense for this local probe, or hold its projection fixed, to avoid adding a new range-finder confound. The old-source contribution is shared and need not be rerun. This is a single-layer algebra test on stored arrays; no Monte Carlo or full-network run is required.

[D] Let r=teacher-baseline, f=(proposed-baseline)/(2-β_0), and g=(naive-proposed)/(4-β_0^2), evaluated separately on off-diagonal D21 and d3. For η=2-β_0 the proposed squared-error reduction is:

```
G_linear = 2 eta <r,f> - eta^2 ||f||^2
```

[D] Record this unnormalized gain, the relative error, and the naive gain. The comparisons use fixed theoretical coefficients; do not fit a scalar to the target. [S] If the proposed variant improves but the naive variant does not, and g accounts for the difference, that supports the confound explanation. If both behave alike or the proposed variant loses across captures, it does not explain those frozen one-step errors. This does not by itself rule out a cumulative effect over many births. An end-to-end run on held-out networks must decide that separate question.

[E] The local test costs several ordinary matrix contractions per capture, reusing the existing birth/slice routines, rather than a full estimator trajectory. Its exact probe cost depends on which transported factors are cached; the deployment ledger above does not count the offline test.

What the data cannot decide; d3 drift

Limits of the diagnostic and a principled calibration target

No decisive global ranking from diagonal κ_4 alone

[D] At the level of moment information, a symmetric perturbation ΔQ supported on permutations of (i,i,j,k), with i,j,k distinct, leaves every Q_{aaaa}, C and T unchanged but changes the first-order response on page 1. Thus the listed teacher statistics do not determine mixed-κ_4 feedback. Knowing W defines the actual network distribution, but evaluating these additional moments still requires new computation.

[D] A minimal extra diagnostic is a directional fourth-cumulant statistic. For a selected vector v, collect:

```
q_i(v,v) = E[u_i^2 (u.v)^2]
           - C_ii (v.T C v) - 2(C v)_i^2
q_i(v,w) = E[u_i^2 (u.v)(u.w)]
           - C_ii (v.T C w) - 2(C v)_i(C w)_i
```

[D] With v_a=diag(d)W[:,a], the leading mixed-κ_4 contribution to the next slice is:

```
Delta_Q D21_ac = (1/2) sum_i e_i [
 W_ic q_i(v_a,v_a) + 2 W_ia q_i(v_a,v_c)]
```

[D] These quantities, with the corresponding regenerated-Q response subtracted, directly test the κ_4 hypothesis. They cannot be recovered from Q diagonal, C and the κ_3 slice. A small output-coordinate subset is sufficient for a directional falsifier, but requires extra stored fourth-moment contractions or new samples. [S] I would request those before designing another carried κ_4 object for this κ_3 problem.

A norm deficit is not a conservation law

[D using Q] The reported L16 norm ratio a=0.986 and relative d3 error e=0.046 imply, treating the quoted teacher as reference:

```
cos(theta) = (a^2+1-e^2)/(2a) = 0.99902637
beta_LS = (a^2+1-e^2)/(2a^2) = 1.01321133
beta_norm = 1/a = 1.01419878
min relative error after scalar fitting = 0.04411705
```

[D] Even the best oracle scalar lowers this d3 squared error by only about 8.02%, not most of it. The norm deficit is not evidence that every coordinate needs the same gain. There is no layerwise conservation law fixing ||d3|| through arbitrary W and ReLU. Both d3 and diag(D21) describe the same cumulant; rescaling one without the other makes the state inconsistent.

[D] A production scalar is principled only relative to an independently specified calibration loss. For the immediate mean channel, the Edgeworth sensitivity is $j_i=-\alpha_i\varphi(\alpha_i)/(6\sigma_i^2)$. A local weighted calibration would set:

```
beta = sum_i j_i^2 d3_chain_i d3_reference_i
       / sum_i j_i^2 d3_chain_i^2
```

[D] The reference must come from independent calibration data or an independently validated moment estimate. This formula is only for the local mean channel; a final-mean objective needs the downstream Jacobian and other coupled channels. [S] Do not hard-code 1.014 from one teacher network. First verify the existing birth compensation already matches its univariate post-ReLU third-moment closure; if it does, scalar renormalization is an empirical calibration, not a missing mathematical identity.

[Q] Sources: attached request of 10 October 2026 (all experimental numbers); read-only `submissions/v55As125/estimator.py`, especially lines 1542-1563 and its source-readout contractions; `experiments/probes/r7_fbscale_run.py`. Primary background: Wu et al., §4.3 and the paper and technical supplement. The paper distinguishes the diagram approximation from its factorized representation; it does not certify the proposed correction. The algebra and cost ledger here are derived independently.